Communication Networks and Services Performance Assignment

Final Phase

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Outline

Project Overview

Goal: Evaluate placement and routing algorithms in a scenario built on YAFS.

Core Strategy Implementation:

- **Routing:** Minimize Latency, Maximize Bandwidth, Random Path
- **Placement:** Minimize Execution Time, Minimize Resource Usage, Random Node

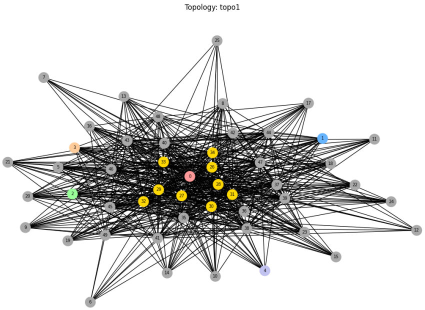
Workload Characteristics:

- **Traffic Models:** Exponential ($\lambda = 4$) & Uniform [1, 5]
- **Resilience test:** Dynamic node failure injected at $t = 10\,000$

Experimental Phase Matrix

Topology	Placement	Routing
BA Topology	MinExecTime	MinLatency
	MinExecTime	MinLatency
	MinExecTime	MaxBW
	MinResUsage	MinLatency
	Random	Random
Grid 2D	MinExecTime	MinLatency
	MinExecTime	MinLatency
	MinExecTime	MaxBW
	MinResUsage	MinLatency
	Random	Random

Topology 1 – Barabási-Albert (50 nodes, $m = 25$)



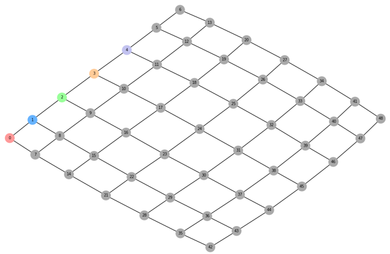
Structural Attributes

- Scale-free model with power-law degree property
- Highly dense core cluster (hub-and-spoke behavior)
- **Gateways (GW):** Nodes 0–4 **Cloud:** Node 49
- **Fog Infrastructure:** Nodes 5–48 (Yellow: dominant hubs)
- **Link params:** PR $= 2$, BW $= 75\,000$
- **Capacities:** IPT $= 1000$, RAM $= 8192$

Key Network Behavior

Extremely short average path steps due to high hub density. This structurally minimizes default transmission latency.

Topology 2 – Grid 2D ($7 \times 7 = 49$ nodes)



Structural Attributes

- Uniform mesh structure; internal nodes share a degree of 4
- Complete absence of high-degree clusters or central hubs
- Balanced and uniform data path dispersion
- **Gateways (GW):** Nodes 0–4 (Edge locations)
- **Fog Layer:** Nodes 5–47 **Cloud Link:** Node 48
- Maintains identical resource parameters as BA

Key Network Behavior

Longer multi-hop paths. Algorithmic path-finding decisions have a much higher impact here than in BA.

Application Architecture & Failure Execution

Complex Workflow (App0–App4)

Every deployed app enforces a non-linear **fork/join pattern**:

- Source $\longrightarrow$ COMP1 (Splitting Node)
- Branch A: COMP1 $\longrightarrow$ COMP2 $\longrightarrow$ Sink1
- Branch B: COMP1 $\longrightarrow$ COMP3 $\longrightarrow$ COMP4 $\longrightarrow$ Sink2
- 4 computing modules: RAM $= 1024$ each
- Data unit size $= 15\,000$ bytes per module
- Processing weight $= 200$ instructions/module

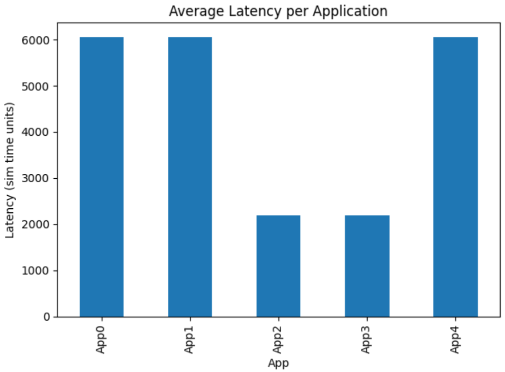
Dynamic Resilient Failure Model

- Abrupt removal of a target compute node at $t = 10\,000$
- Target choice: lowest-degree exposed processing node
- Seamless dynamic failover execution inside `get_path_from_failure()`
- Active cache invalidation forces paths to bypass the fault zone
- Total observation horizon extended until $t = 20\,000$

Validation Strategy

Validates real-time algorithm survivability and routing adaptability when links alter mid-session.

Topo 1 – Latency: MinExecTime + MinLatency (Exponential)



Key Observations

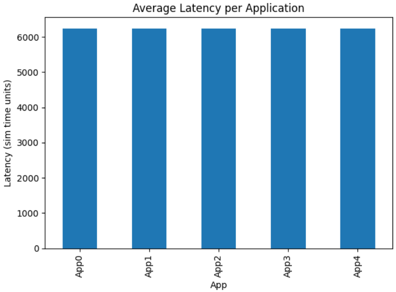
- Average system latency matches ≈ **4 000** sim time
- Unified and low variance across all 5 apps
- Result of identical optimal structural choices (highest IPT node selected)
- Low delay variation verifies routing predictability

Infrastructure Bottleneck

All operational modules target the same high-capacity Cloud/hub engine. No local cluster distribution happens.

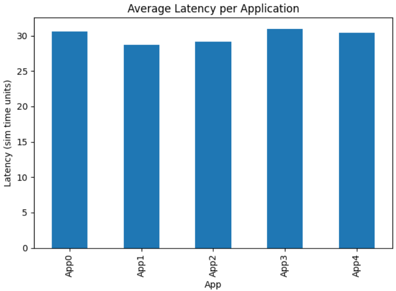
Topo 1 – Latency: MinResUsage + MinLatency vs Random

MinResUsage + MinLatency



Mean Latency ≈ **6 200** sim units

Random Node + Random Path



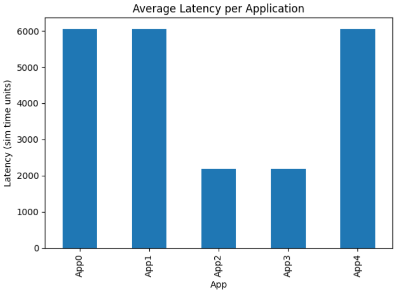
Mean Latency ≈ **30** sim units

Algorithmic Insights – Why Random Achieves Lower Latency

Random placement disperses tasks close to source endpoints, avoiding large multi-hop pipelines. Conversely, MinResUsage picks massive RAM hosts located deeper across the core, extending default routing distances.

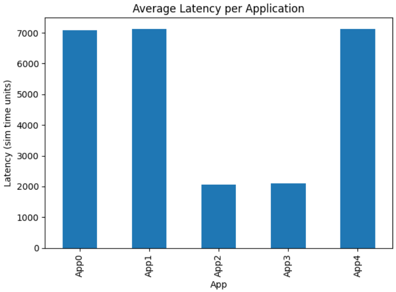
Topo 1 – Traffic Distribution Impact (Exponential vs Uniform)

Exponential Model ($\lambda = 4$)



Mean Latency $\approx$ **4 000**

Uniform Model [1, 5]



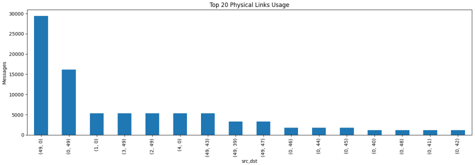
Mean Latency $\approx$ **4 200**

Traffic Dynamics Explanation

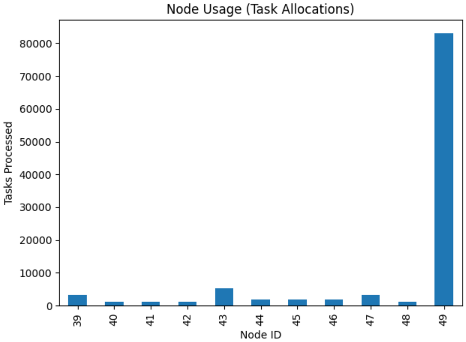
The Uniform strategy triggers higher packet arrival concurrency (103 378 vs 89 084 entries, +16%). This behavior saturates shared lines, lifting processing latency by $\sim 4.5\%$.

Topo 1 – Link & Node Utilization Breakdown

Top-20 Link Usage Allocation



Node Allocation Density

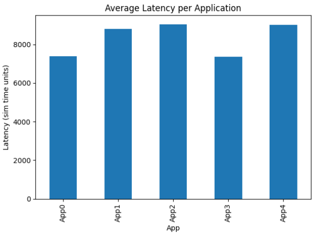


Analytical Deduction

A distinct minority of backbone links route the massive bulk of system packets, reflecting classic scale-free graph behaviors. Resource profiles cluster heavily around a single high-performance target.

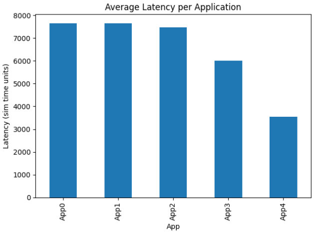
Topo 2 – Latency Analysis Across Strategies

MinExecTime + MinLat



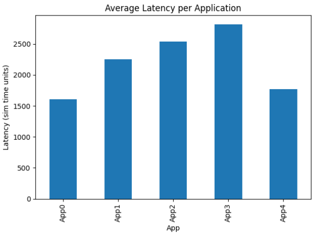
Latency ≈ **8 100**

MinResUsage + MinLat



Latency ≈ **5 500**

Random + Random



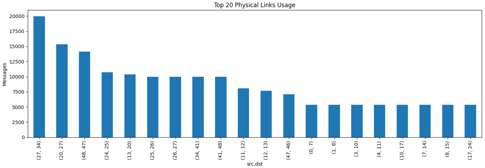
Latency ≈ **2 100**

Mesh Topology Penalizes Extreme Centralization

MinExecTime forces allocation onto a remote peripheral node (Cloud Node 48). Lacking central hubs, this increases hop penalties significantly compared to the scale-free BA layout.

Topo 2 – Link & Node Utilization Breakdown

Top-20 Line Traffic Allocation



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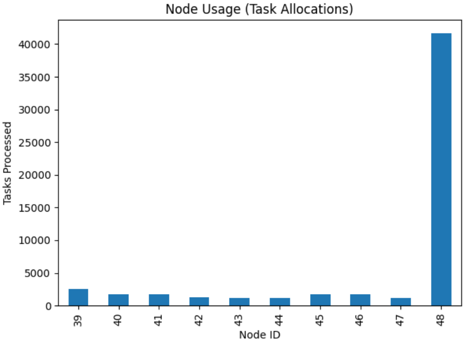


Analytical Deduction

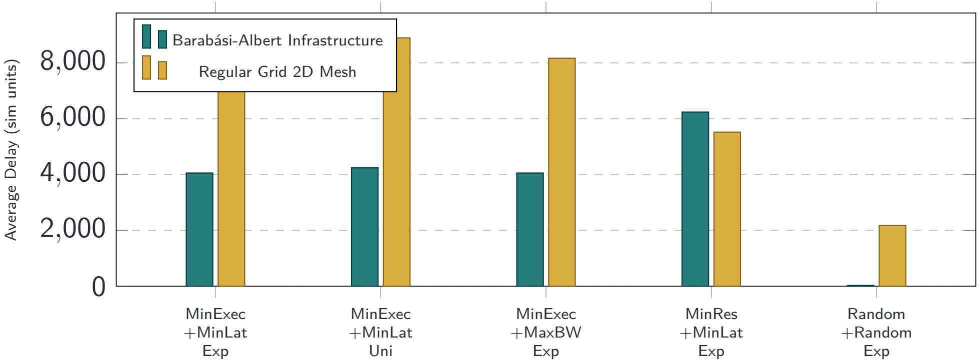
Traffic flow spreads smoothly across adjacent routes without a single dominant link hotspot. The core bottleneck shifts to the routing paths leading to the remote Cloud cluster.

Consolidated Performance Metrics

Network Shape	Placement Rules	Routing Model	Traffic	Mean Latency
Barabási-Albert (T1)	MinExecTime	MinLatency	Exp	4 056
Barabási-Albert (T1)	MinExecTime	MinLatency	Uni	4 241
Barabási-Albert (T1)	MinExecTime	MaxBandwidth	Exp	4 056
Barabási-Albert (T1)	MinResUsage	MinLatency	Exp	6 238
Barabási-Albert (T1)	**Random Node**	**Random Path**	Exp	**30**
Grid 2D Mesh (T2)	MinExecTime	MinLatency	Exp	8 166
Grid 2D Mesh (T2)	MinExecTime	MinLatency	Uni	8 895
Grid 2D Mesh (T2)	MinExecTime	MaxBandwidth	Exp	8 166
Grid 2D Mesh (T2)	**MinResUsage**	**MinLatency**	Exp	**5 520**
Grid 2D Mesh (T2)	Random Node	Random Path	Exp	2 176

Metric Convergence Fact: MinLatency and MaxBandwidth yield matching delays here. Because link properties are uniform across the network configuration, path cost minimizing filters match identical shortest route chains.

Cross-Topology Comparative Data Visualization



Conclusions

Topology Impact

- Centralized scale-free networks naturally limit routing penalties through well-connected cluster backbones.
- Grid mesh models heavily penalize centralized setups by increasing multi-hop penalties (up to $\times 2$ delay inflation).

Routing Convergence

MinLatency aligns with MaxBandwidth in environments with uniform link weights. Distinct paths only emerge when handling heterogeneous link parameters.

Placement Strategic Priority

- Grouping processing jobs on a isolated engine type creates systematic bottlenecks.
- Distributed setups (Random, MinResUsage) optimize routing paths by keeping processing closer to sources.
- Smarter future approaches must utilize path-aware logic.

Traffic Distribution Effects

Uniform distributions generate $+15\%$ higher packet concurrency, inflating system latency across both network topologies.

Thank You

Questions?

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